

## Assignment Questions of RDBMS (BCA 2B)

### Q1. University Admission Management System

A university wants to automate its admission process. Each student:

- Can apply for multiple programs.
- Each program belongs to one department.
- Each department has multiple faculty members.
- One faculty member is the Head of the department.

#### Tasks:

- a) Identify entities and attributes.
- b) Draw ER diagram (mention cardinalities).
- c) Convert ER diagram into relational tables with primary and foreign keys.
- d) Write SQL queries to:
  - Insert 5 student records
  - Retrieve students who applied for more than one program

### Q2. Online Shopping Portal

An e-commerce company maintains:

- Customers
- Products
- Orders
- Payments

#### Rules:

- A customer can place multiple orders.
- Each order can contain multiple products.
- Payment is done per order.
- Products belong to categories.

#### Tasks:

- a) Identify relationships and specify their type (1:1, 1:M, M:N).
- b) Design normalized relational tables (up to 3NF).
- c) Write SQL queries to:
  - Find total order amount per customer
  - Display top 3 most sold products
- d) Explain what type of index should be created and why.

### Q3. Hospital Database System

A hospital stores:

- Doctors (specialization)
- Patients
- Appointments
- Rooms

#### Rules:

- A patient can consult multiple doctors.
- A doctor can treat many patients.
- Each appointment is scheduled in one room.

#### Tasks:

- a) Explain why M:N relationship arises and how you resolve it.
- b) Design relational schema with constraints.
- c) Write SQL queries:

- List doctors who handled more than 10 patients
- Find room utilization count
- d) Explain what happens internally when an index is created on doctor\_id.

#### **Q4. Banking System**

A bank maintains:

- Customers
- Accounts (Savings/Current)
- Transactions
- Branches

Rules:

- A customer can have multiple accounts.
- Each account belongs to one branch.
- Transactions are linked to accounts.

**Tasks:**

- Identify strong and weak entities.
- Design relational schema with ON DELETE CASCADE where appropriate.
- Write SQL queries to:
  - Find customers with balance > ₹50,000
  - Display branch-wise total deposits
- Explain low cardinality vs high cardinality attributes in this database.

#### **Q5. Railway Reservation System**

The railway system maintains:

- Trains
- Stations
- Passengers
- Tickets

Rules:

- A train runs between multiple stations.
- A passenger can book multiple tickets.
- Ticket contains seat number and journey date.

**Tasks:**

- Identify composite and candidate keys.
- Design relational schema with proper constraints.
- Write SQL queries to:
  - Find passengers travelling on a specific date
  - Count passengers per train
- Explain how B+ Tree indexing helps in ticket search by journey date.